Machine Learning · 3rd-Year Software Engineering

Chapter 2 — Data Preparation & Exploratory Data Analysis

Session 2 of 2 · Encoding, scaling, correlation & EDA, data leakage

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Warm-up: three questions from session 1

RETRIEVAL · 4 min · individual, no notes

Answer on paper.

1 · Mean and population standard deviation of (2, 2, 6, 6)?

2 · A fare value is missing in a right-skewed column. Mean or median to fill it? Why?

3 · Write the upper fence of the IQR rule.

Retrieval without notes strengthens memory more than re-reading.

Warm-up correction

#	Answer	Reasoning
1	$\bar{x} = 4$, $\sigma = 2$	Deviations $-2, -2, 2, 2 \rightarrow$ squares sum 16 $\rightarrow \sigma^2 = 16/4 = 4$
2	Median	Robust to extreme values (Titanic fare: mean 32.20, median 14.45)
3	$Q_3 + 1.5 \cdot IQR$	With $IQR = Q_3 - Q_1$

By the end of this session, you can do four things

Each one is practised and checked in the final assessment

1. Encode nominal and ordinal variables without inventing an order
2. Standardize or min–max scale a feature, and choose by context
3. Compute covariance and Pearson's r, and read EDA results as hypotheses
4. Explain data leakage and prevent it: split first, fit on train, transform both

Session 1 recap

- Data types
- Mean, variance, std
- Missing values
- Outliers (IQR)

PART 1

Encode categories into numbers

A model cannot compute "Cherbourg − Southampton"

One-hot encoding: one binary column per category, no order invented

Nominal feature embarked $\in$ {C, Q, S}

Passenger	embarked	embarked_C	embarked_Q	embarked_S
1	S	0	0	1
2	C	1	0	0
3	Q	0	1	0
4	S	0	0	1

Properties

k categories $\longrightarrow$ k columns

Exactly one 1 per row

All categories are equally far from each other

Same column names as scikit-learn's OneHotEncoder (checked)

Ordinal encoding: integer codes that preserve a real order

Ordinal feature size $\in \{S < M < L\}$

size	size_code
S	0
M	1
L	2

What it assumes

An order exists: S < M < L ✓

For a linear model, equal steps too: S→M counts as much as M→L

Titanic class (1st/2nd/3rd) is ordered; the chapter's pipeline one-hot encodes it, which avoids assuming equal steps — both choices are defensible

Three encoding errors that quietly mislead a model

✘ Ordinal codes on a nominal feature

C = 0, Q = 1, S = 2 tells a linear model that Q lies between C and S, and that S − C = 2 × (Q − C). None of this is true.

✘ One-hot on thousands of categories

Thousands of postal codes → thousands of sparse columns. Group rare categories or use another strategy (later chapters).

⚠ Unseen category at prediction

A category absent from training appears in production. In the chapter's pipeline, handle_unknown="ignore" encodes it as all zeros.

The encoding is an assumption about the variable: one-hot = no order; ordinal = order (and, for linear models, equal steps)

Your turn: choose an encoding

ACTIVITY · 2 min · pairs

Which encoding for each feature?

1 · Day of the week (Mon … Sun)

2 · Postal code (thousands of distinct values)

3 · Education level (primary < secondary < higher)

4 · Payment method (card / cash / mobile)

Fingers: 1 = one-hot · 2 = ordinal · 3 = neither directly (needs another strategy)

Correction: the variable's meaning decides the encoding

Feature	Encoding	Why
Day of the week	One-hot (1)	Its order is cyclic (Sunday → Monday): integers 0–6 put Sunday far from Monday
Postal code	Neither directly (3)	High cardinality: group values (e.g. by region) or use another strategy
Education level	Ordinal (2)	A real order exists
Payment method	One-hot (1)	Nominal, few categories

Cyclic encodings (e.g. sine/cosine of the day) exist for periodic features — an advanced option, not needed today

PART 2

Put numerical features on comparable scales

Standardization and min–max normalization

Without scaling, merely changing a unit changes the nearest neighbour

Synthetic passengers: A (22 years, 7.25) · B (60 years, 8.05) · C (23 years, 70.00). Euclidean distance from A:

Age expressed in…	d(A, B)	d(A, C)	Nearest
years	38.0	62.8	**B**
months	456.0	63.9	**C**

Same passengers, same information — only the unit of age changed.

Lesson

The feature with the largest numeric range dominates the distance — not the most important one.

Sensitive: kNN, k-means, SVM, gradient descent. Not sensitive: decision trees.

Two common rescalings, both learned from the training data

Standardization (z-score)

$$z_i = (x_i - \mu) / \sigma$$

Result: mean 0, standard deviation 1; no fixed range

Min–max normalization

$$x'_i = (x_i - x_{min}) / (x_{max} - x_{min})$$

Result: range [0, 1] on the training data

Fitted statistics

μ, σ, x_{min}, x_{max} are computed from data → computed on the training split only.

Test values may then fall outside [0, 1] — that is expected.

scikit-learn: StandardScaler (z-score, population σ) and MinMaxScaler — each with fit on train, transform on both

Worked example: standardizing and normalizing three ages

Ages: 20, 30, 40 → $\mu = 30$, $\sigma = \sqrt{(200 / 3)} \approx 8.165$

	20	30	40
z-score (x − 30) / 8.165	−1.22	0	1.22
min–max (x − 20) / 20	0	0.5	1

Check: the z-scores average to 0 ✓ and the min–max values span [0, 1] ✓

Reading a z-score

z = 1.22 means 1.22 standard deviations above the mean.

The unit (years) disappears.

One extreme value can squash every other standardized value

Synthetic fares: 8, 9, 10, 250 → μ = 69.25, σ ≈ 104.36

	8	9	10	250
z-score	−0.59	−0.58	−0.57	**1.73**
log, then z-score	−0.66	−0.57	−0.50	**1.73**

z-score: the three ordinary fares end up within 0.02 of each other.

log first: their spread is about 8 times larger (0.15) — the model can still tell them apart.

Options

Log transform (positive data; use log(1 + x) if zeros)

Robust scaling (median and IQR)

Cap the extreme value first

Your turn: which preparation would you choose?

ACTIVITY · 2 min · pairs

Choose a scaling (or none) and justify it.

a · Pixel intensities between 0 and 255

b · Household income, strongly right-skewed, as input to a kNN model

c · Numerical features of a decision tree

Two pairs present their answer; the others agree or object.

Correction: the data and the model decide the scaling

a · Pixels → min–max
Known, fixed bounds (0–255): dividing by 255 maps to [0, 1].

b · Income → log(1 + x), then standardize
The log reduces the skew; standardization then puts income on the same footing as other features for distances.

c · Tree → no scaling needed
A tree splits on thresholds of one feature at a time; a monotonic rescaling does not change the possible splits.

Scaling is not a ritual: apply it when the model uses distances or gradients, and choose the formula from the data's shape

PART 3

Measure relationships and explore the data

Covariance, Pearson's r, and reading EDA results as hypotheses

Covariance gives the direction of a relation; Pearson's r gives its strength

Covariance

$$\mathrm{Cov}(x, y) = (1/n) \cdot \Sigma\,(x_i - \bar{x})(y_i - \bar{y})$$

> 0: move together · < 0: opposite · depends on the units

Pearson correlation

$$r = \mathrm{Cov}(x, y) \,/\, (\sigma_x \cdot \sigma_y) \quad \in [-1, 1]$$

Unit-free: ±1 perfect linear relation · 0 no linear relation

Read r carefully

r measures linear association only.

Correlation is not causation.

Worked example: covariance and r for five points

$x = (1, 2, 3, 4, 5)$, $y = (2, 4, 5, 4, 5) \rightarrow \bar{x} = 3, \bar{y} = 4$

						Σ
$x_i - \bar{x}$	−2	−1	0	1	2	**0**
$y_i - \bar{y}$	−2	0	1	0	1	**0**
product	4	0	0	0	2	**6**

Cov = 6 / 5 = **1.2** $\sigma_x = \sqrt{2} \approx 1.414$ $\sigma_y = \sqrt{1.2} \approx 1.095$

r = 1.2 / (1.414 × 1.095) ≈ **0.77** → fairly strong positive linear relation (NumPy np.corrcoef: 0.7746 ✓)

r = 0 does not mean "unrelated", and r ≠ 0 does not mean "causes"

Counter-example: $y = x^2$

$x = (-2, -1, 0, 1, 2) \rightarrow y = (4, 1, 0, 1, 4)$

r = 0 exactly, yet y is completely determined by x.

The relation is real but not linear.

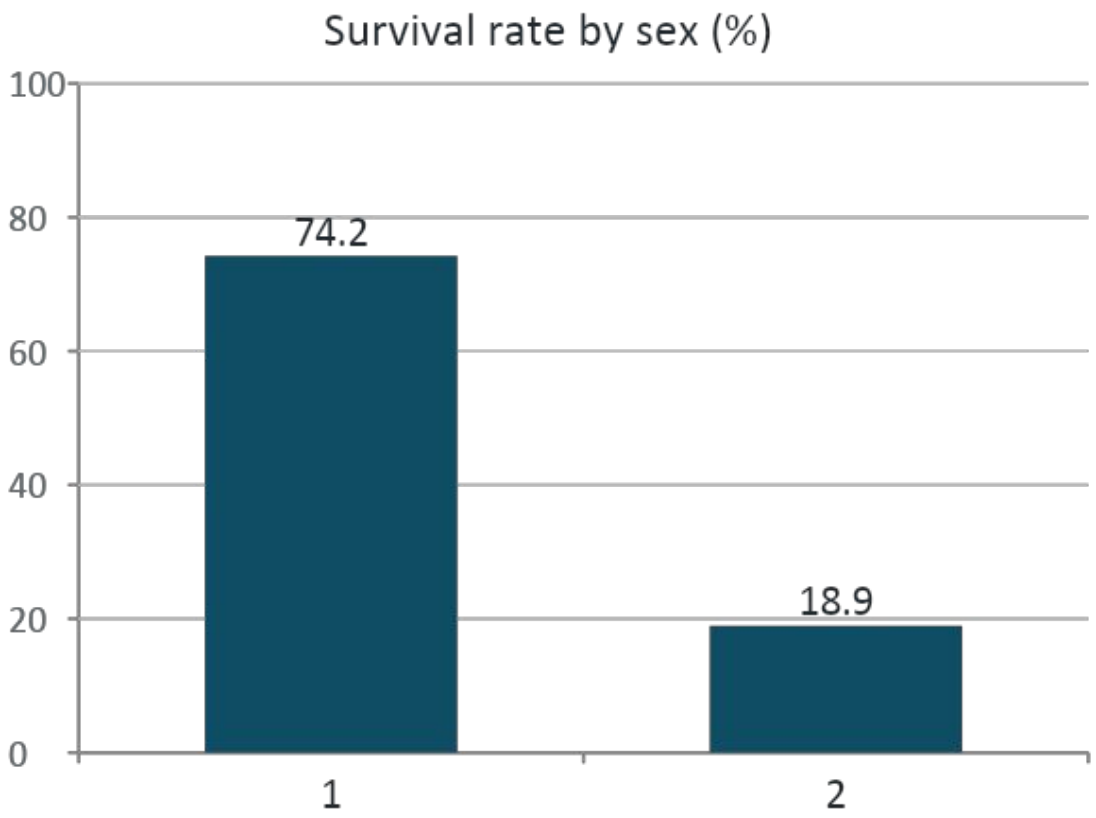
Correlation ≠ causation

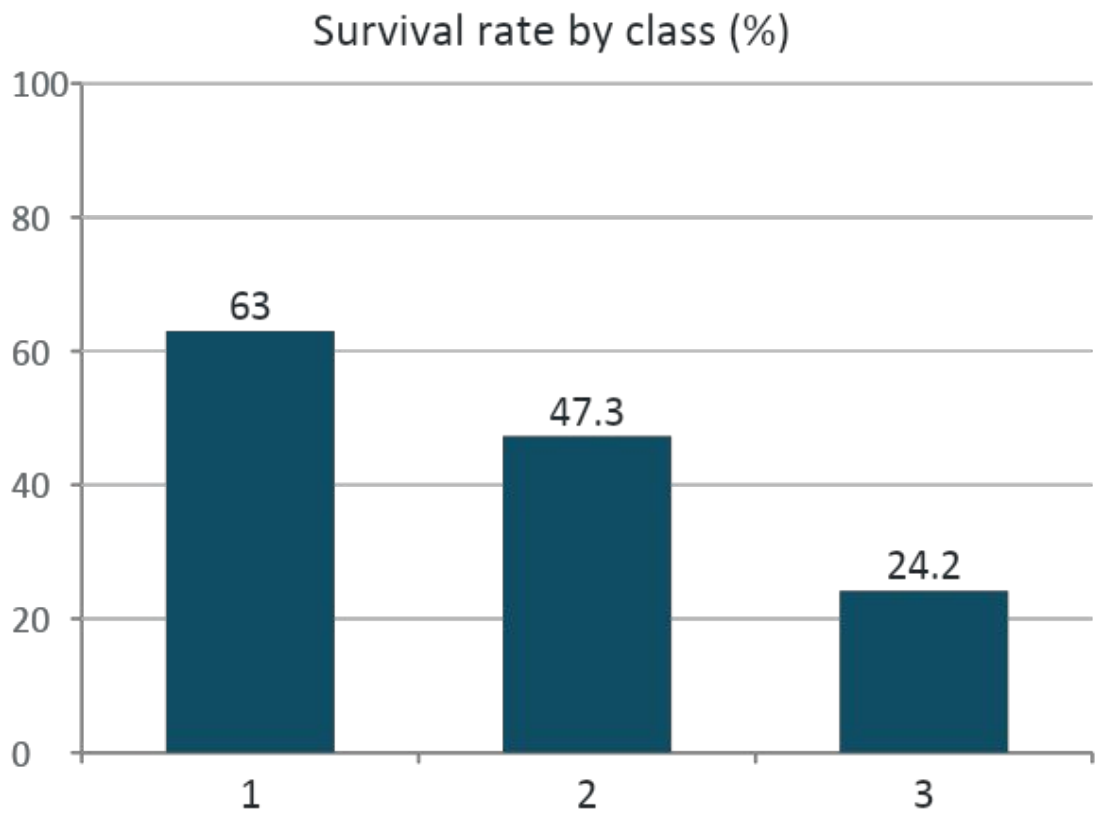
Two features can move together because a third factor drives both, or by coincidence.

EDA findings are hypotheses to test, not conclusions.

Before trusting a correlation, plot the data: a scatter plot reveals curves that r misses

EDA on Titanic: survival differs sharply by sex and by class





Overall survival: 38.4% (342 / 891). These gaps are real signal in this dataset — and a fairness question for a later chapter

Discuss: does paying a higher fare cause survival?

Pearson r with survived

Feature	r
fare	0.26
parch	0.08
age	−0.08
sibsp	−0.04

Pairs · 2 min

fare has the largest correlation with survival among these numeric features.

Can we conclude that a more expensive ticket caused survival? What else could explain this correlation?

Median fare by class: 1st 60.29 · 2nd 14.25 · 3rd 8.05

PART 4

Prevent data leakage

The rule that keeps every number of this chapter honest

Data leakage: information unavailable at prediction time sneaks into training

- Definition: any information not available at prediction time — especially test-set statistics — that influences training
- Effect: the evaluation becomes over-optimistic
- Three frequent forms:
  - test rows used to compute a preprocessing statistic
  - a feature that contains the target in disguise
  - modelling choices made by looking at the test set

Consequence

An excellent test score…

…followed by disappointing results in production — because the test was not a fair rehearsal.

On Titanic, including the test rows changes the fill value for age

29.0

median age computed on the training split

28.0

median age computed on all 891 rows (train + test)

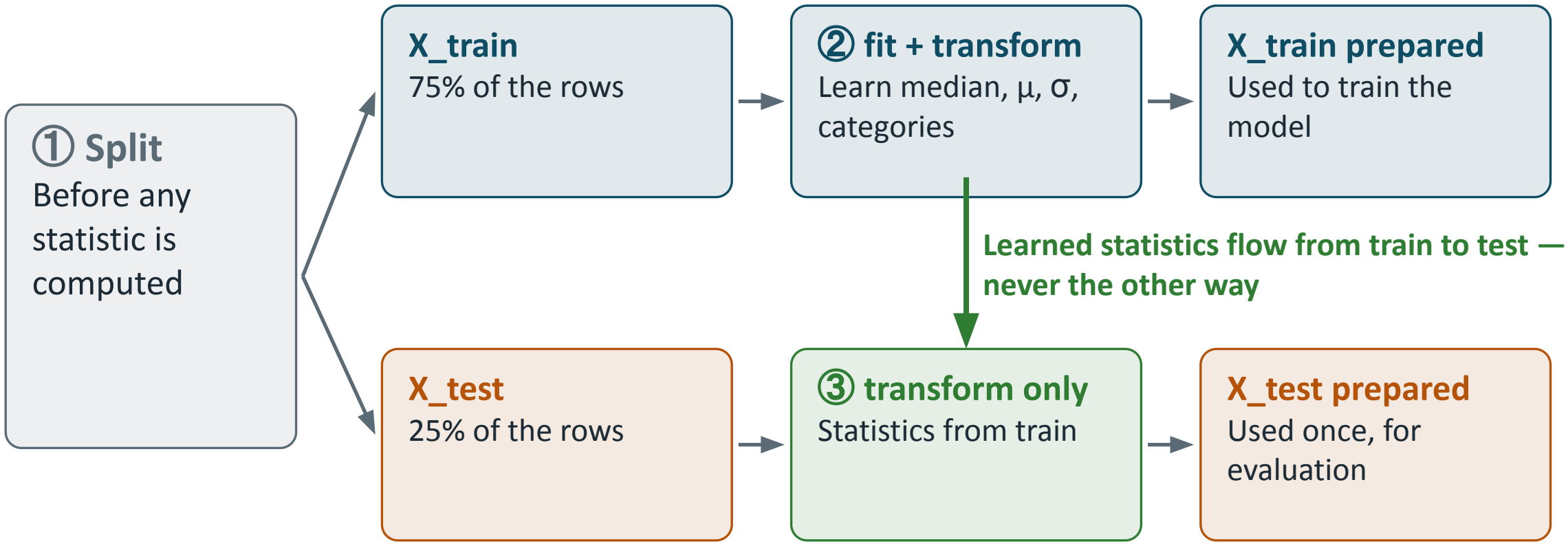
Setting

Chapter split: 75 / 25, stratified on survived, random_state = 42 (executed)

Meaning

Small shift here, but the test rows did influence a learned value. The same holds for every fitted statistic: means, σ, categories.

The leakage-safe order: split first, fit on train, transform both



The test set never influences a fitted statistic: this is exactly what fit_transform(X_train) then transform(X_test) enforces

Spot the bug: this code looks fine but leaks

ACTIVITY · 2 min · pairs

Find the line that leaks and explain the consequence.

```
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.25, random_state=42)
```

Raise your hand when your pair has found the line number (1–7).

Correction: split first, then fit the scaler on the training set only

```
# BUG (line 5): fit_transform(X) used all rows:
# μ and σ contain test-set information.

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=42)
scaler = StandardScaler()
# fit on the training set only
X_train_s = scaler.fit_transform(X_train)
# transform only on the test set
X_test_s = scaler.transform(X_test)
```

In the lab

Pipeline and ColumnTransformer bundle imputation, encoding and scaling into one fitted object, so the right order is the default.

Full treatment: Chapter 12.

Leakage also hides in features and in habits

Target in disguise

The seaborn Titanic file contains a column alive (yes/no).

It equals survived on all 891 rows (checked) → must be removed before training.

Peeking at the test set

Exploring the test set and letting what you see shape your choices is leakage too.

Do EDA on the training split.

Training–serving skew

Production must apply the same fitted preprocessing.

Save the fitted preprocessor with the model; validate inputs (unknown categories, ranges).

Ask of every feature and every statistic: would I know this at prediction time, from training data only?

Final assessment: four questions

ASSESSMENT · 4 min · individual

Q1. You fit StandardScaler on the full dataset, then split. Consequence?

A · The test score may be over-optimistic

B · The model cannot be trained

C · Nothing changes

Q2. Best encoding for embarked (C / Q / S) in a linear model?

A · Ordinal codes 0 / 1 / 2

B · One-hot

C · Drop the column

Q3. Pearson's r between two features is 0.02. You can conclude…

A · they are unrelated

B · there is no linear relation; a non-linear one is possible

C · one does not cause the other

Q4. Which must be computed on the training split only?

A · The median used for imputation

B · The scaler's mean and σ

C · Both A and B

Show 1, 2 or 3 fingers for A, B or C, question by question, on the count of 3.

Final assessment correction

Q	Answer	Why
1	A	The scaler's μ and σ used test rows: leakage, optimistic evaluation
2	B	embarked is nominal: ordinal codes would invent an order
3	B	r measures linear association only (recall $y = x^2$, $r = 0$)
4	C	Every fitted statistic is learned on the training split, then applied to the test

What to remember from Chapter 2

- Split first, fit on the training set, transform both — for every learned statistic
- One-hot for nominal categories; ordinal codes only for a real order
- Scale for distance- and gradient-based models; log or robust scaling when outliers dominate
- Pearson's r: linear association only, and never a proof of causation
- Leakage also hides in features (alive) and in habits (peeking at the test set)

Next: the lab

- Audit and prepare the real Titanic file in pandas
- Build the ColumnTransformer pipeline
- Measure the leakage gap (advanced exercise 1)

References and data used in this chapter

- Kaufman, S., Rosset, S., Perlich, C., & Stitelman, O. (2012). Leakage in data mining: Formulation, detection, and avoidance. ACM Transactions on Knowledge Discovery from Data. doi:10.1145/2382577.2382579
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- Géron, A. Hands-On Machine Learning with Scikit-Learn and PyTorch. O'Reilly (2025 edition — confirm the year of the copy you cite)
- Tukey, J. W. (1977). Exploratory Data Analysis. Addison-Wesley
- Data: titanic.csv from the seaborn-data repository (github.com/mwaskom/seaborn-data), 891 rows × 15 columns

Figures recomputed by executing Python (NumPy 2.5.3, pandas 3.0.5, scikit-learn 1.9.1). Synthetic examples are labelled as such.